

Md Mahmuddun Nabi Murad

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Research Profile

Ph.D. Candidate in Electrical Engineering at the University of South Florida specializing in deep learning for time-series forecasting and anomaly detection. First-author papers at **ICML 2026** and **AAAI 2025**, applied time-series machine learning experience at **Schneider Electric**, and winner in a **global Hackathon**. Experienced with PyTorch, Transformers, MLP-Mixers, wavelet-based modeling, and signal-processing-informed approaches.

Technical Skills

Languages: Python, C/C++

ML / Data: PyTorch, scikit-learn, NumPy, Pandas; TensorFlow (familiar)

Methods: Time-series forecasting, anomaly detection, Transformers, MLP-Mixers, diffusion models, RNNs, wavelet-based modeling

Tools: Git, Linux, Optuna, MATLAB, Databricks, LaTeX

Experience

Time Series Machine Learning Intern

Jun 2026 – Aug 2026

Schneider Electric, Andover, MA

- Developed a global **anomaly detection model** to detect anomalies in the UPS (Uninterrupted Power Supply).

Graduate Research Assistant

Aug 2023 – Present

SISLab, University of South Florida

- Developing efficient deep learning architectures for **time-series analysis**, **environmental monitoring**, and **intelligent marine transportation systems**.
- Conducting research leading to first-author publications at top conferences, including **ICML** and **AAAI**.

Protection Engineer (Data & Systems Analysis)

Dec 2019 – Jul 2023

Power Grid Bangladesh PLC

- Fault analysis, relay coordination, high-voltage testing and commissioning, and automation configuration.

Education

Ph.D. in Electrical Engineering (ML Focus)

University of South Florida (USF), Tampa, FL

CGPA: 4.0/4.0

Aug 2023 – Present

M.S. in Electrical Engineering (ML Focus)

University of South Florida (USF), Tampa, FL

CGPA: 4.0/4.0

Aug 2023 – May 2026

B.Sc. in Electrical and Electronic Engineering

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

CGPA: 3.67/4.0

Feb 2015 – Apr 2019

SELECTED MACHINE LEARNING RESEARCH

- **Online Time-Series Anomaly Detection:** Designed a sequential anomaly detection framework using a causal mixer and cluster-based embeddings in the reconstruction model; published as first author at ICML 2026.
- **Long-Term Time-Series Forecasting:** Designed a wavelet-based decomposition and reconstruction model for multi-variate forecasting and reported improvements over state-of-the-art variants; published as first author at AAAI 2025.
- **Vessel Trajectory Prediction:** Developed a hybrid Transformer-MLP-Mixer architecture for AIS-based vessel trajectory prediction and a custom route endpoint detection method; the work secured 1st Place globally in the 2024 M3 Hackathon.
- **AIS and Vision-Derived Maritime Modeling:** Developed a pipeline for vessel-route prediction from AIS or vision-derived data and curated a maritime video dataset for benchmarking in complex real-world visual environments; published as first author at Ocean Engineering Journal.
- **Environmental Monitoring:** Estimated soil moisture and localized sensors from cumulative received RF signal strength for landslide detection, and estimated snow depth from weather data as an alternative to expensive snow stations. Published the sensor localization work as first author at IEEE-WCNC 2025.
- **Outlier Crew Detection and Work-hour Estimation:** Designed a framework for outlier person detection among crew members on a ship, along with work hour estimation; published as first author at MaCvi 2025.

Research Publications

- [1] M. M. N. Murad and Y. Yilmaz, "Cluster-Aware Causal Mixer for Online Anomaly Detection in Multivariate Time Series," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
- [2] M. M. N. Murad, M. Aktukmak, and Y. Yilmaz, "Wpmixer: Efficient multi-resolution mixing for long-term time series forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, 2025.
- [3] M. M. N. Murad, B. S. Turgut, and Y. Yilmaz, "Mix&Fix-Net: A Dual-Stage Trajectory Prediction Model for AIS and Vision-Derived Vessel Data," *arXiv preprint arXiv:2608.19580*, 2026, **Ocean Engineering**.
- [4] M. M. N. Murad, D. K. Yirenya-Tawiah, T. Weller, and Y. Yilmaz, "Multi-Resolution Mixer Network for Localization of Multiple Sensors From Cumulative Power Measurements," in *IEEE Wireless Communications and Networking Conference (WCNC)*, 2025.
- [5] M. M. N. Murad, B. S. Turgut, A. Ahmed, G. Camliyurt, and Y. Yilmaz, "Camera-based Intruder Detection and Monitoring of Ship Crew Work Hours," in *Proceedings of the Winter Conference on Applications of Computer Vision*, 2025.

Awards and Prizes

Managing Maritime Movements (M3) Hackathon

Winner (First Prize), 2024

(Organized by the U.S. Department of Homeland Security.)

M3 is a global hackathon on small vessel route prediction. I developed a novel Transformer–MLP-Mixer hybrid model for route prediction and proposed a route's endpoint detection method.

BUET Dean's list award

Achieved the prestigious dean's list award for outstanding academic performance in one semester during undergraduate studies.